OMER ABDELGADIR MOHAMED YASSIN

JUNIOR ELECTRONICS ENGINEER

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Objective

Detail-oriented and motivated Electronics Engineering graduate with a strong academic foundation in embedded systems, microcontroller programming, and circuit design. Demonstrated hands-on project experience building IoT-based applications, integrating sensors, and designing PCBs using KiCad and LTspice. Strong understanding of core communication protocols and hardware debugging. Eager to bring technical proficiency and analytical problem-solving skills to an entry-level engineering role to drive reliable hardware and software solution

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Education

BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING
FUTURE UNIVERSIRTY, **SUDAN**
GRADUATION DATE: MARCH 2026

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skills

technical skills

- **Core Engineering:** Circuit Design & Analysis, Signal Processing & Communication Systems, Sensors & Actuators Integration
- **Microcontrollers & Hardware:** Arduino, ESP32
- **PCB & Simulation Tools:** KiCad, Proteus, LTspice
- **Programming & Protocols:** C, C++, Python, I2C, SPI, UART, Wi-Fi, Bluetooth

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COURSES AND TRAINING

- **Semiconductor Fabrication 101 Certificate** – Purdue University
- **Mastering KiCad: Open-Source PCB Design for Beginners** – Coursera Instructor Network
- **Linear Circuits 1: DC Analysis** – Georgia Institute of Technology
- **Linear Circuits 2: AC Analysis** – Georgia Institute of Technology

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PROJECTS

Sonar-based Object Detection System

- Engineered an IoT-based sonar detection platform utilizing Arduino microcontrollers and ultrasonic sensors to measure distances up to 4 meters with an accuracy of ± 1 cm.
- Developed optimized C++ firmware to process real-time sensor streams and display telemetry data on an LCD interface
- Performed comprehensive system validation and environmental testing across varying ambient conditions, reducing false-positive detection rates by 15%

Extracurricular Activities

Electronics Club Member| FUTURE UNIVERSITY (2020-2026)

- Collaborated with cross-functional student teams on embedded systems and IoT hardware developments
- Mentored over 15 junior students in core Arduino programming concepts and basic circuit design methodologies
- Contributed to the design, prototyping, and testing phases of 3+ electronic prototypes, including health monitoring systems and home automation units
- Authored technical documentation and step-by-step project guides to facilitate skill-building workshops for club members

Portfolio website : omer-abdelgadir.github.io/personal-portfolio